

NORTH SOUTH UNIVERSITY

Bashundhara, Dhaka, Bangladesh

TruthLens AR

Bangla Multimodal Fake News, Image & Deepfake Detection

Using Vision-Language Models

COURSE INFORMATION:

Department	Electronics & Communication Engineering (ECE)
Semester	Summer 2026
Course Code	CSE 499A-Senior Design Project I
	CSE 499B-Senior Design Project II

SUBMITTED BY:

STUDENT NAME	STUDENT ID
Subroto Ghosh	2212150642
Joy Saha	2312466042
MD. Abu Sufian Protik	2312234042
Anas Ahmed	2121264042

SUBMITTED TO

Mohammad Shifat-E-Rabbi (MSRB)
Assistant Professor, North South University

Date of Submission: June 24, 2026

ABSTRACT

Misinformation spreads rapidly across digital platforms, posing significant threats to public discourse and societal trust. Current detection systems address individual modalities (text, image, or video) separately, lacking unified detection capabilities and explainability. This proposal introduces TruthLens, a comprehensive Vision-Language Model-based system for detecting and explaining misinformation across multiple modalities. The proposed system integrates text analysis, image verification, and deepfake detection into a single framework while providing natural language explanations for predictions. We present a four-month implementation plan including model development, cross-platform validation, and evaluation against baseline methods. Expected outcomes include a deployable system, peer-reviewed publication, and open-source contributions. This work addresses the critical need for explainable, unified misinformation detection in the AI community.

I. INTRODUCTION

The proliferation of false information across social media platforms represents one of the most pressing challenges in the digital age. According to recent studies, misinformation spreads six times faster than factual information on social networks. Fake news articles, manipulated images, and deepfake videos collectively undermine public trust and can influence significant outcomes ranging from personal decisions to political elections.

Current detection approaches suffer from fundamental limitations. Text-based systems analyze articles in isolation. Image forensics tools focus solely on visual manipulation. Deepfake detectors target video artifacts. In reality, misinformation integrates multiple modalities simultaneously. A fake news article accompanied by doctored images creates a more convincing narrative than individual components. Furthermore, existing systems typically provide binary classifications without explaining their reasoning, limiting user understanding and trust.

Recent advances in Vision-Language Models have demonstrated exceptional capabilities in cross-modal understanding. Models such as CLIP and LLaVA can process and relate textual and visual information simultaneously. This proposal leverages these advances to develop TruthLens, an integrated system for detecting misinformation across modalities while providing interpretable explanations for predictions.

II. PROBLEM STATEMENT

A. Lack of Unified Detection Systems

Most detection frameworks handle single modalities. A text-based model cannot verify accompanying images. An image forensics tool cannot assess textual claims. This fragmentation requires users to employ multiple tools and cross-reference results, creating friction and reducing adoption.

B. Limited Explainability

Existing systems produce verdicts without justification. When a model flags content as fake, users cannot understand why. This opacity undermines confidence and prevents meaningful verification by non-experts.

C. Cross-Platform Generalization

Models trained on one social platform often fail on others due to distribution shifts. Robust cross-domain performance remains an open challenge.

D. Real-Time Deployment Constraints

Vision-Language Models are computationally expensive. Current approaches are too slow for real-time social media monitoring, limiting practical applicability.

III. PROJECT OBJECTIVES

Primary Objective: Develop TruthLens, a unified Vision-Language Model-based system that detects and explains misinformation across text, image, and video modalities with high accuracy and cross-platform robustness.

Objective 1

Create a single Vision-Language Model that processes text, images, and video simultaneously, eliminating the need for separate tools.

Objective 2

Implement an explanation module generating natural language reasoning for predictions, enabling users to understand detection decisions.

Objective 3

Develop a confidence scoring mechanism (0-100 scale) replacing binary classifications to capture nuanced credibility assessments.

Objective 4

Validate system performance across multiple platforms and content types to ensure generalization beyond training data distribution.

IV. PROPOSED APPROACH

Our solution integrates state-of-the-art Vision-Language Models with domain-specific preprocessing and post-processing modules to create a comprehensive misinformation detection pipeline.

A. System Architecture

- 1) **Input Processing:** Text normalization, image resizing and augmentation, video frame extraction and sampling ensure consistent input dimensions and formats across modalities.
- 2) **Vision-Language Feature Extraction:** CLIP or LLaVA processes normalized inputs, producing joint embeddings that capture both visual and semantic information in a unified vector space.
- 3) **Multimodal Fusion:** Attention-based fusion mechanisms combine extracted features, modeling relationships between text and visual elements to detect inconsistencies characteristic of misinformation.
- 4) **Classification Head:** Fully connected layers with dropout regularization map fused features to probability distributions over credibility classes. Softmax outputs generate confidence scores.
- 5) **Explanation Generator:** Fine-tuned GPT-2 or T5 model generates natural language explanations by conditioning on attention weights and feature importance scores from the classification head.

B. Dataset Strategy

We will construct a multi-source dataset combining: (1) Textual misinformation from LIAR and FakeNewsNet; (2) Manipulated images from established forensics benchmarks; (3) Deepfake videos from FaceForensics++ and Celeb-DF; (4) Social media content from Twitter, Reddit, and TikTok. Annotations will follow standard credibility labels with inter-annotator agreement verification.

V. IMPLEMENTATION PLAN

Phase	Duration	Key Activities
Phase 1	Month 1	Literature review, dataset assembly, annotation
Phase 2	Month 2	Baseline development, BERT and ResNet integration
Phase 3	Month 3	VLM integration, explanation module, dashboard UI
Phase 4	Month 4	Cross-domain validation, benchmarking, documentation

VI. EVALUATION METRICS

Performance Metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC across multiple test sets. We will compare against established baselines including individual modality-specific models.

Robustness Evaluation: Cross-domain testing by training on one platform and evaluating on others. Adversarial robustness against intentional manipulation attempts.

Explainability Assessment: User studies evaluating whether generated explanations improve human judgment. Interpretability metrics measuring feature attribution validity.

Computational Efficiency: Inference latency, memory requirements, and throughput on standard hardware to assess deployment feasibility.

VII. EXPECTED OUTCOMES

Deliverable 1: Functional TruthLens system deployable as a web application enabling users to submit content for credibility assessment and receive confidence scores with explanations.

Deliverable 2: Pre-trained model weights and source code released on GitHub with comprehensive documentation, enabling reproducibility and community contributions.

Deliverable 3: Peer-reviewed research paper submitted to premier venues presenting novel architecture and empirical results demonstrating state-of-the-art performance.

Deliverable 4: Technical documentation, API specification, and deployment guides facilitating adoption by researchers and practitioners.

VIII. SIGNIFICANCE AND IMPACT

This project addresses a critical societal challenge while advancing state-of-the-art in multimodal learning. By developing an explainable, unified detection system, we contribute to combating misinformation at scale. The open-source release enables researchers worldwide to build upon our work. Industry adoption potential spans social media platforms, fact-checking organizations, and news outlets. Educational value derives from demonstrating best practices in Vision-Language Model integration and deployment.

IX. NOVELTY AND CONTRIBUTION

1. Unified Multimodal Framework: First integrated system handling text, image, and video misinformation detection within a single Vision-Language Model.

2. Interpretable Detection: Integration of explanation generation modules providing natural language reasoning for predictions, advancing explainable AI in content verification.
3. Cross-Domain Robustness: Systematic evaluation across multiple platforms demonstrating generalization capabilities beyond training distribution.
4. Practical Deployment: Complete system including optimization strategies for real-time inference, bridging research and production systems.

X. REFERENCES

- [1] Wang, W. Y., et al. (2017). Liar, liar pants on fire: A new benchmark dataset for fake news detection. In Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), pp. 422-426.
- [2] Shu, K., et al. (2019). FakeNewsNet: A data repository and news challenge for improved fact checking and misinformation detection. In Conference on Empirical Methods in Natural Language Processing (EMNLP).
- [3] Radford, A., et al. (2021). Learning transferable visual models from natural language supervision. In International Conference on Machine Learning (ICML), pp. 8748-8763.
- [4] Liu, H., et al. (2023). LLaVA: Large Language and Vision Assistant. arXiv preprint arXiv:2304.08485.
- [5] Li, Y., et al. (2019). FaceForensics++: Learning to detect manipulated facial images. In IEEE International Conference on Computer Vision (ICCV), pp. 1-11.
- [6] Devlin, J., et al. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT).
- [7] He, K., et al. (2016). Deep residual learning for image recognition. In IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 770-778.
- [8] Raffel, C., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. Journal of Machine Learning Research (JMLR), 21(140):1-67.
- [9] Simonyan, K., Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. In International Conference on Learning Representations (ICLR).
- [10] Goodfellow, I., et al. (2014). Generative adversarial nets. In Advances in Neural Information Processing Systems (NIPS), pp. 2672-2680.
- [11] Kingma, D. P., Ba, J. (2014). Adam: A method for stochastic optimization. In International Conference on Learning Representations (ICLR).
- [12] McInnes, L., Healy, J., Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for dimension reduction. arXiv preprint arXiv:1802.03426.